BTS-I(R/S)/II(S)-11.18-1904 Reg. No. [ ] **B**

B.Tech. Degree I Semester Regular and Supplementary/ II Semester Supplementary Examination November 2018

CE/EE GE 15-1203 A ENGINEERING GRAPHICS

(2015 Scheme)

Time: 3 Hours Maximum Marks: 60

$(5 \times 12 = 60)$

I. An area of 50 km^2 of a field is represented by an area of 150 cm^2 on a map. Determine the R.F. of the scale used in the map. Also construct a diagonal scale to show kilometers, hectometers and decimeters. The maximum length to be indicated on the scale is 10 km. Show a distance of 6.48 km on the scale.

OR

II. The asymptotes of a hyperbola are at an angle of $75°$. A point P on the curve is 25 mm away from the horizontal asymptote and 40 mm from the inclined asymptote, when measured horizontally. Construct the curve and draw tangent and normal at a point M on the curve, 50 mm above the horizontal asymptote.

III. A pipeline from a point A, running due south east, has a downward gradient of one in five. Another pipeline from point B running S 20° E meets the pipeline from A at C. If the point B is in the same level with and 15 m due east of point A, find the slope and true length of each pipeline.

OR

IV. Draw the projections of a hexagonal lamina of 36 mm side resting with one of its corners on HP such that the diagonal through that corner is inclined at $35°$ to VP and the surface of the lamina is inclined at $40°$ to HP.

V. A cylindrical disc of 70 mm diameter. and 20 mm thickness has a square hole of 30 mm side, cut centrally through its flat face. Draw the projections of the disc, when its flat faces are inclined at 30° to VP and any two rectangular faces of the hole are inclined $25°$ to HP.

OR

VI. An octahedron of 32 mm edges, rests on one of its faces, with an edge of face perpendicular to VP. It is cut by a sectional plane inclined at 30° to HP and passing through the centre of gravity of the solid. Draw the sectional plan.

VII. A vertical cylinder, base 90 mm diameter is penetrated by another horizontal cylinder of 50 mm diameter. The axis of the horizontal cylinder is 5 mm infront of the axis of the vertical cylinder. Draw the curves of intersection.

OR

VIII. Draw the development of the lateral surface of a right regular pentagonal prism of 30 mm base edge and 70 mm height, kept vertically on HP. An ant moves in a straight line from one base corner, at an angle of θ° to HP always and reaches the top corner vertically above it after rounding the lateral surface once. Draw the path traced out by the ant in the front view and measure the angle θ.

IX. A frustum of a cone, base dia. 32 mm, top face dia. 24 mm and length of axis 36 mm, is resting centrally on the frustum of a hexagonal pyramid, base 40 mm side, top face 30 mm and height 24 mm. Draw the isometric projection of the compound solid.

OR

X. A nut in the form of a hexagonal prism, side of base 3.2 cm and axis 3.2 cm with a cylindrical hole through it, 3.2 cm dia. the axis of hole coinciding with the axis of the prism. The base of the prism is on the ground and the nearest vertical edge is in the picture plane. The face containing the nearest edge is inclined at 10° to the picture plane. The height of the station point is opposite to the centre of the nut and its distance from the picture plane is 15 cm. Draw a perspective of the nut.

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